

Yuan Tuo

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Education

2020–26	Ph.D., Economics, University of California, Davis (expected)
2018–19	B.Ec., Economics (First Class Honours), The University of Sydney
2017	Exchange, University of Wisconsin–Madison
2015–18	B.Ec., Economics, The Australian National University
2014–15	Wuhan University, China

Research Interest

Econometrics, Applied Econometrics

Working Papers

Optimal Treatment Under Spillover Effects and Noncompliance (Job Market Paper)

Abstract: This paper considers optimal treatment allocation in the presence of spillover effects in both compliance and outcomes. I show that when there are two layers of spillover effects, an individual's outcome depends on other individuals' compliance types in the neighborhood. By empirically maximizing welfare subject to a budget constraint or minimizing cost subject to a welfare goal, I propose an optimal randomized treatment rule based on the characteristics of the target population. I show that the proposed treatment rule achieves asymptotic welfare efficiency with an $n^{-1/2}$ convergence rate for the welfare loss, and is asymptotically weakly feasible. I also apply the proposed treatment rule to an anti-conflict experiment in public middle schools in New Jersey and show that not incorporating both layers of spillover effects can result in misallocation of treatment.

Causal Inference in RCTs with Multiple Treatment Arms

Abstract: When there is more than one treatment group in an RCT with noncompliance, local average treatment effects (LATE) are not comparable because each treatment has different compliers. I propose a partial-identification method to compare two treatments in the presence of noncompliance. In particular, I derive sharp bounds on the average difference between pairs of treatment effects for those who would comply with both treatments, as well as for other subpopulations. I also apply the bounds derived to an experiment evaluating the impacts of different depression treatments on economic outcomes in India.

Work in Progress

Inference of Marginal Treatment Effect (MTE) Estimator (with Takuya Ura)

References

Takuya Ura

UC Davis
Associate professor
takura@ucdavis.edu

Shu Shen

UC Davis
Associate professor
shushen@ucdavis.edu

Colin Cameron

UC Davis
Professor
accameron@ucdavis.edu

Presentations

- 2025 California Econometrics Conference
- 2025 Midwest Econometrics Group Conference

Fellowships and Awards

- 2023 Dean's Graduate Summer Support Award, University of California, Davis
- 2021 First-year Fellowship, University of California, Davis

Teaching Experience

- 2020–25 **Teaching Assistant, University of California, Davis**
Undergraduate
Analysis of Economic Data, Econometrics, Economic & Financial Forecasting,
Economics & Business Data Analytics, Principles of Microeconomics,
Principles of Macroeconomics, Intermediate Macro Theory, Public Finance
Graduate (Ph.D. level)
Econometric Methods, Cross-Sectional Econometrics
- 2019 **Tutor, The University of Sydney**
Introductory Microeconomics, Introductory Macroeconomics

Referee Service

Econometrics Review

Software Skills

R, Stata, Matlab

Languages

Mandarin (native), English (fluent), Japanese (JLPT N1)